Practical No. 3

Title: Write a program to solve a fractional Knapsack problem using a greedy method.

```
def fractional_knapsack(W, wt, val, n):
  items = sorted(
    zip(wt, val),
    key=lambda x: x[1] / x[0],
    reverse=True
  )

  total = 0

  for w, v in items:
    if W == 0:
      break
    if w <= W:
      total += v
      W -= w
    else:
      total += v * W / w
      W = 0

  return total

n = int(input("Enter number of items: "))
val = list(map(int, input("Enter values: ").split()))
wt = list(map(int, input("Enter weights: ").split()))
W = int(input("Enter knapsack capacity: "))

print("Maximum value:", fractional_knapsack(W, wt, val, n))
```

Output:

```
PS C:\Users\HP\AppData\Local\Programs\Microsoft VS Code> & C:\Users\HP\AppData\Local\Python\py
thoncore-3.14-64\python.exe c:/Users/HP/Downloads/3_Knapsack.py
Enter number of items: 2
Enter values: 22 4
Enter weights: 32 2
Enter knapsack capacity: 3
Maximum value: 4.6875
PS C:\Users\HP\AppData\Local\Programs\Microsoft VS Code>
```